AI Knowledge Base Dataset

102 simple documents in 6 categories for your search project

How to use this file

Each document below has an ID, a category and a title, followed by its text. Each one is short and written in simple words. In your project, read the text, split it by document ID, and keep the category and title with each document.

The company policies and training details are sample content. You can change them to your own.

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Python

DOC-001 | CATEGORY: PYTHON

What Is Python

Python is an easy programming language. People use it for websites, data work and AI. Its code is simple to read.

DOC-002 | CATEGORY: PYTHON

Variables and Data Types

A variable is a name that holds a value, like age = 25. Common types are int (whole numbers), float (decimals), str (text) and bool (True or False).

DOC-003 | CATEGORY: PYTHON

Python Lists

A list holds many items in order, like [1, 2, 3]. You can add, remove and sort items. The first item is at position 0.

DOC-004 | CATEGORY: PYTHON

Python Dictionaries

A dictionary stores pairs of a key and a value, like {'name': 'Ravi', 'age': 30}. You find a value by using its key. It is good for records and settings.

DOC-005 | CATEGORY: PYTHON

Tuples and Sets

A tuple is a list that cannot be changed. A set keeps only unique items and removes repeats. Use a tuple for fixed data and a set to remove duplicates.

DOC-006 | CATEGORY: PYTHON

If Statements and Conditions

An if statement lets the program choose. The code runs only when the condition is true. Use elif for another check and else for everything else.

DOC-007 | CATEGORY: PYTHON

For and While Loops

A for loop repeats code for each item in a list. A while loop repeats as long as a condition is true. Use break to stop a loop early.

DOC-008 | CATEGORY: PYTHON

Defining Functions

A function is a block of code you can use again and again. You create it with def. It can take inputs and give back a result with return.

DOC-009 | CATEGORY: PYTHON

Modules and Imports

A module is a Python file with code you can reuse. You bring it in with import, like import math. Python also has many extra packages you can install.

DOC-010 | CATEGORY: PYTHON

Reading and Writing Files

Python can open a file with the open function. Use 'r' to read and 'w' to write. The with statement closes the file for you.

DOC-011 | CATEGORY: PYTHON

Exception Handling

Errors can stop your program. Put risky code in a try block and handle the error in an except block. The program then keeps running.

DOC-012 | CATEGORY: PYTHON

Classes and Objects

A class is a plan for making objects. An object holds data and actions together. The __init__ method sets up each new object.

DOC-013 | CATEGORY: PYTHON

List Comprehensions

A list comprehension makes a new list in one short line, like [x * 2 for x in numbers]. You can add a condition to pick only some items. It is shorter than a loop.

DOC-014 | CATEGORY: PYTHON

Virtual Environments and pip

A virtual environment keeps each project's packages separate. This avoids version problems. Use pip to install packages, like pip install numpy.

DOC-015 | CATEGORY: PYTHON

String Methods

Strings have helpful tools. lower and upper change letter case, strip removes extra spaces, split breaks text into words and replace swaps text. They are used to clean text.

DOC-016 | CATEGORY: PYTHON

Lambda Functions

A lambda is a tiny function written in one line, like lambda x: x + 1. Use it when you need a quick function once. For longer code, use def.

DOC-017 | CATEGORY: PYTHON

Working with JSON Data

JSON is a common way to share data between programs. Python's json module turns JSON into dictionaries and lists. It is how you read data from web services.

AI/ML

DOC-018 | CATEGORY: AI/ML

AI, Machine Learning and Deep Learning

AI means making machines act smart. Machine learning is AI that learns from data. Deep learning is machine learning that uses big neural networks.

DOC-019 | CATEGORY: AI/ML

Supervised Learning

In supervised learning, the model learns from examples with correct answers. For example, emails marked spam or not spam. It is used to predict categories or numbers.

DOC-020 | CATEGORY: AI/ML

Unsupervised Learning

Unsupervised learning finds patterns in data with no answers given. Clustering groups similar items together. An example is grouping customers by buying habits.

DOC-021 | CATEGORY: AI/ML

Reinforcement Learning

In reinforcement learning, a program learns by trying actions. It gets rewards for good actions and penalties for bad ones. It is used in games and robots.

DOC-022 | CATEGORY: AI/ML

Train and Test Split

We split data into two parts. The training part teaches the model and the test part checks it. A common split is 80 percent training and 20 percent testing.

DOC-023 | CATEGORY: AI/ML

Overfitting

Overfitting means the model remembers the training data too well. It then does badly on new data. More data and simpler models can fix this.

DOC-024 | CATEGORY: AI/ML

Underfitting

Underfitting means the model is too simple to learn the pattern. It does badly on both training and new data. A stronger model or better features can help.

DOC-025 | CATEGORY: AI/ML

Linear Regression

Linear regression predicts a number using a straight line. For example, it can guess a house price from its size. It is a simple model to start with.

DOC-026 | CATEGORY: AI/ML

Logistic Regression

Logistic regression is used to predict a category, like yes or no. It gives a chance between 0 and 1. A cut-off, often 0.5, makes the final choice.

DOC-027 | CATEGORY: AI/ML

Decision Trees

A decision tree asks a series of yes or no questions, like a flowchart. It is easy to understand. One tree can overfit, so trees are often combined.

DOC-028 | CATEGORY: AI/ML

Random Forest

A random forest is many decision trees working together. Each tree votes and the most popular answer wins. It is more accurate than one tree.

DOC-029 | CATEGORY: AI/ML

Neural Networks

A neural network is made of layers of small connected units called neurons. It learns by adjusting the strength of the connections. It can learn from images, text and sound.

DOC-030 | CATEGORY: AI/ML

Gradient Descent

Gradient descent is how a model learns. It changes its settings little by little to make fewer mistakes. Think of walking downhill to reach the lowest point.

DOC-031 | CATEGORY: AI/ML

Accuracy, Precision and Recall

Accuracy is how many predictions are right overall. Precision is how many predicted positives are truly positive. Recall is how many real positives were found.

DOC-032 | CATEGORY: AI/ML

Text Embeddings

An embedding is a list of numbers that shows the meaning of text. Texts with similar meaning get similar numbers. This lets computers search by meaning, not just words.

DOC-033 | CATEGORY: AI/ML

Cosine Similarity

Cosine similarity tells how close two embeddings are. A score near 1 means very similar and near 0 means not related. It is used to rank search results.

DOC-034 | CATEGORY: AI/ML

Retrieval-Augmented Generation (RAG)

RAG first searches for the best documents for a question. Then an AI writes the answer using those documents. This makes answers more correct.

NumPy

DOC-035 | CATEGORY: NUMPY

What Is NumPy

NumPy is a Python library for working with numbers. Its main tool is the array. Arrays are faster than normal Python lists.

DOC-036 | CATEGORY: NUMPY

Creating NumPy Arrays

Make an array with np.array, like np.array([1, 2, 3]). A list inside a list makes a table with rows and columns. All items in an array have the same type.

DOC-037 | CATEGORY: NUMPY

Array Shape and Reshape

The shape tells how many rows and columns an array has, like (3, 4). Reshape changes the shape without changing the values. The total number of items must stay the same.

DOC-038 | CATEGORY: NUMPY

NumPy Data Types

Every array has a data type, like int32 or float64. Check it with dtype and change it with astype. A smaller type uses less memory.

DOC-039 | CATEGORY: NUMPY

Array Indexing

Use square brackets to get an item, starting from 0. For tables use row and column, like arr[1, 2]. A negative number counts from the end.

DOC-040 | CATEGORY: NUMPY

Array Slicing

Slicing gets a range of items, like arr[1:4]. It returns positions 1, 2 and 3. Use arr[:, 0] to get the first column.

DOC-041 | CATEGORY: NUMPY

Boolean Masking

A mask picks values by a condition. For example, arr[arr > 10] gives only values above 10. It is fast and needs no loop.

DOC-042 | CATEGORY: NUMPY

Broadcasting

Broadcasting lets arrays of different sizes work together. For example, adding one number to a whole array adds it to every item. It keeps code short.

DOC-043 | CATEGORY: NUMPY

Arithmetic on Arrays

Operations like +, -, * and / work on every item at once. NumPy also has np.sqrt, np.exp and np.log. This is much faster than a loop.

DOC-044 | CATEGORY: NUMPY

Aggregation Functions

Aggregation functions give one summary number. Examples are sum, mean, median, min, max and std. Use axis to choose rows or columns.

DOC-045 | CATEGORY: NUMPY

zeros, ones, arange and linspace

np.zeros and np.ones make arrays full of 0 or 1. np.arange makes numbers with a fixed step. np.linspace makes a set number of evenly spaced values.

DOC-046 | CATEGORY: NUMPY

Random Numbers in NumPy

np.random makes random numbers. For example, random.rand gives values between 0 and 1. Set a seed to get the same numbers every time.

DOC-047 | CATEGORY: NUMPY

Dot Product and Matrix Multiplication

The dot product multiplies matching items and adds them up to give one number. Use np.dot or the @ sign. It is the base of similarity scores.

DOC-048 | CATEGORY: NUMPY

Transpose of an Array

Transpose swaps rows and columns. A shape of (2, 3) becomes (3, 2). Use arr.T.

DOC-049 | CATEGORY: NUMPY

Joining Arrays

np.concatenate joins arrays together. np.vstack stacks them up and down, and np.hstack puts them side by side. The shapes must match.

DOC-050 | CATEGORY: NUMPY

Vectorization and Speed

Vectorization means working on a whole array at once instead of using a loop. It can be many times faster. It is a key habit in data science.

DOC-051 | CATEGORY: NUMPY

Vector Norm and Normalization

The norm is the length of a vector. Use np.linalg.norm to find it. Normalizing makes the length 1, so cosine similarity becomes a simple dot product.

Pandas

DOC-052 | CATEGORY: PANDAS

What Is Pandas

Pandas is a Python library for table data, like Excel or CSV files. Its two main parts are the Series and the DataFrame. It is used to clean and study data.

DOC-053 | CATEGORY: PANDAS

Pandas Series

A Series is one column of data with labels. It works like a list but is faster and smarter. Each column in a DataFrame is a Series.

DOC-054 | CATEGORY: PANDAS

Pandas DataFrame

A DataFrame is a table with rows and columns. It is like an Excel sheet. You can make one from a dictionary or a file.

DOC-055 | CATEGORY: PANDAS

Reading CSV Files

Use pd.read_csv to load a CSV file into a DataFrame. You can choose columns and the separator. Pandas can also read Excel and JSON files.

DOC-056 | CATEGORY: PANDAS

Inspecting a DataFrame

Use head to see the first rows. Use info to see column types and missing values. Use describe to see simple statistics.

DOC-057 | CATEGORY: PANDAS

Selecting Columns

Use df['price'] to get one column. Use df[['name', 'price']] to get several columns. Column names are case sensitive.

DOC-058 | CATEGORY: PANDAS

loc and iloc

loc picks rows and columns by label. iloc picks them by position number. For example, df.iloc[0] gives the first row.

DOC-059 | CATEGORY: PANDAS

Filtering Rows

Put a condition in brackets to filter rows, like df[df['age'] > 30]. Use & for and, and | for or. Put each condition in parentheses.

DOC-060 | CATEGORY: PANDAS

Handling Missing Values

Missing data shows as NaN. Use isna to find it, dropna to remove it and fillna to fill it with a value. Choose based on how much is missing.

DOC-061 | CATEGORY: PANDAS

Removing Duplicates

Duplicate rows can give wrong results. Use duplicated to find them and drop_duplicates to remove them. Use subset to check only some columns.

DOC-062 | CATEGORY: PANDAS

Adding and Changing Columns

Make a new column by assigning to it, like df['total'] = df['price'] * df['qty']. Use rename to change names and drop to remove columns.

DOC-063 | CATEGORY: PANDAS

Sorting Data

Use sort_values to order rows by a column. It goes from small to large by default. Use ascending=False for large to small.

DOC-064 | CATEGORY: PANDAS

GroupBy

groupby splits data into groups and does a calculation on each. An example is the average salary for each department. It is like GROUP BY in SQL.

DOC-065 | CATEGORY: PANDAS

Merging DataFrames

merge joins two tables using a shared column. It works like a SQL join. You can choose inner, left, right or outer.

DOC-066 | CATEGORY: PANDAS

Using apply

apply runs your own function on every item in a column. It is good for custom cleaning. For simple maths, built-in operations are faster.

DOC-067 | CATEGORY: PANDAS

Pivot Tables

A pivot table summarizes data with rows, columns and a calculated value. In Pandas use pivot_table. It is like a pivot table in Excel.

DOC-068 | CATEGORY: PANDAS

Saving Data with to_csv

Use df.to_csv('file.csv', index=False) to save a DataFrame as a CSV file. Pandas can also save Excel and JSON files. Saving lets you reuse cleaned data.

Company policies

DOC-069 | CATEGORY: COMPANY POLICIES

Annual Leave Policy

Full-time staff get 20 paid leave days each year. Ask for leave in the HR portal at least two weeks before. Your manager must approve it. You can carry over up to 5 unused days.

DOC-070 | CATEGORY: COMPANY POLICIES

Sick Leave Policy

Staff get 10 paid sick days each year. Tell your manager early on the first day you are sick. A doctor's note is needed if you are off for more than two days in a row.

DOC-071 | CATEGORY: COMPANY POLICIES

Remote Work Policy

You can work from home up to two days a week with your manager's approval. Be available in core hours and use the company VPN. Fully remote work needs written approval from HR.

DOC-072 | CATEGORY: COMPANY POLICIES

Working Hours and Attendance

Work hours are 9:00 AM to 6:00 PM, Monday to Friday, with a one-hour lunch break. Core hours are 10:00 AM to 4:00 PM. Talk to your manager if you are often late.

DOC-073 | CATEGORY: COMPANY POLICIES

Code of Conduct

Everyone must be honest, respectful and professional. Bullying, discrimination and misuse of company resources are not allowed. Report any conflict of interest to HR.

DOC-074 | CATEGORY: COMPANY POLICIES

Data Security and Passwords

Passwords must be at least 12 characters and changed every 90 days. Never share them. Use multi-factor login. Report lost devices or strange emails to IT security.

DOC-075 | CATEGORY: COMPANY POLICIES

Confidentiality Policy

Keep company, client and staff information private, even after you leave. Do not send work files to personal email. Breaking this rule can lead to disciplinary action.

DOC-076 | CATEGORY: COMPANY POLICIES

Dress Code

The normal dress code is smart casual. Wear business formal when meeting clients. Casual clothes are fine on Fridays and at company events.

DOC-077 | CATEGORY: COMPANY POLICIES

Expense Reimbursement

The company pays back approved work costs like travel and client meals. Send your claim within 30 days with original receipts. Claims without receipts may be rejected.

DOC-078 | CATEGORY: COMPANY POLICIES

Anti-Harassment Policy

The company does not accept any harassment. Report problems to HR or the anonymous hotline. Reports are checked quickly and nobody may be punished for reporting.

DOC-079 | CATEGORY: COMPANY POLICIES

Performance Review Process

Reviews happen twice a year, in June and December. They look at goals met, skills growth and feedback. The result can affect promotions, training and pay.

DOC-080 | CATEGORY: COMPANY POLICIES

Overtime Policy

Get your manager's approval before working overtime. Approved overtime is paid at 1.5 times the normal rate or given as time off. Salaried managers do not get overtime pay.

DOC-081 | CATEGORY: COMPANY POLICIES

Public Holidays

The company follows all national public holidays. The holiday list is shared each January. If you work on a public holiday, you get extra pay or a day off.

DOC-082 | CATEGORY: COMPANY POLICIES

Use of Company Equipment

Company laptops and phones are for work. Keep them safe and do not install unapproved software. Return all equipment when you leave the company.

DOC-083 | CATEGORY: COMPANY POLICIES

Business Travel Policy

Get approval from your manager before any business trip. Book economy flights and mid-range hotels through the travel portal. A daily allowance covers meals and local transport.

DOC-084 | CATEGORY: COMPANY POLICIES

Grievance Procedure

If you have a complaint, first tell your manager. If it is not solved, write to HR. HR replies within five working days, and you can appeal to senior management.

DOC-085 | CATEGORY: COMPANY POLICIES

Social Media Policy

Do not share company secrets or speak for the company on social media without permission. Make clear when an opinion is your own. Harmful posts can lead to disciplinary action.

Training information

DOC-086 | CATEGORY: TRAINING INFORMATION

Training Attendance Policy

You must attend every training session. If you miss one, tell your trainer or manager before it starts. Watch the recording within three working days. Missing more than two sessions without a good reason can delay your certificate.

DOC-087 | CATEGORY: TRAINING INFORMATION

Training Program Overview

The 12-week AI and Data Training Program teaches Python up to building an AI project. It has live classes, practice work and a final project. You do not need to know programming before you start.

DOC-088 | CATEGORY: TRAINING INFORMATION

Training Schedule

Live classes are Monday to Thursday, 10:00 AM to 12:00 PM. Afternoons are for self-study and practice. Fridays are for revision, questions and mentoring.

DOC-089 | CATEGORY: TRAINING INFORMATION

Assessments and Quizzes

There is a short quiz every week, a mid-program test in week 6 and a final test in week 12. Each quiz is open for 48 hours and you can try it twice. Your higher score counts.

DOC-090 | CATEGORY: TRAINING INFORMATION

Certification Requirements

To get your certificate you must attend 85 percent of sessions, score at least 60 percent in the final test and submit the final project. Certificates come within two weeks after the program ends.

DOC-091 | CATEGORY: TRAINING INFORMATION

Contacting Your Trainer

Ask questions in the program chat. Trainers reply within one working day. For private matters, email the trainer or book a time on Friday. Add screenshots when you ask about errors.

DOC-092 | CATEGORY: TRAINING INFORMATION

Make-Up Sessions

If you miss a class for a valid reason, like illness, you can ask for a make-up session. Ask the training coordinator within 48 hours of coming back. Make-up sessions are held on Fridays.

DOC-093 | CATEGORY: TRAINING INFORMATION

Accessing the Learning Platform

All course materials, recordings and assignments are on the online learning portal. Log in with your company email and the temporary password from day one. Contact the IT help desk if you cannot log in.

DOC-094 | CATEGORY: TRAINING INFORMATION

Assignment Deadlines

Assignments are given on Monday and due the next Sunday at 11:59 PM. Late work loses 10 percent for each day. After three days it is not accepted unless you had an extension.

DOC-095 | CATEGORY: TRAINING INFORMATION

Capstone Project

In the last four weeks you build a final project, like an AI knowledge base search system. Submit your code, a README and a short presentation. It is graded on correctness, clarity and originality.

DOC-096 | CATEGORY: TRAINING INFORMATION

Training Code of Conduct

Be on time, respectful and active in class. Keep your phone silent during live sessions. Copying someone else's work is plagiarism and can get you removed from the program.

DOC-097 | CATEGORY: TRAINING INFORMATION

Feedback and Evaluation

Trainers give written feedback on graded work within five working days. You also fill in an anonymous survey every month. We use it to improve the training.

DOC-098 | CATEGORY: TRAINING INFORMATION

Prerequisites for Training

You should be able to use a computer, install software and read English instructions. Basic maths like averages helps. A short starter module on the portal covers the basics.

DOC-099 | CATEGORY: TRAINING INFORMATION

Laptop and Software Requirements

Bring a laptop with at least 8 GB of RAM and good internet. Install Python 3.10 or newer, Visual Studio Code and Jupyter Notebook before day one. A setup guide is provided.

DOC-100 | CATEGORY: TRAINING INFORMATION

Session Recordings

Every live class is recorded and posted on the portal within 24 hours. Recordings stay available for 90 days after the program. They are for your own learning only.

DOC-101 | CATEGORY: TRAINING INFORMATION

Grading System

Your final grade is: weekly quizzes 20 percent, assignments 30 percent, mid-program test 15 percent, final test 20 percent and final project 15 percent. Grades are Distinction, Merit, Pass or Not Yet Passed.

DOC-102 | CATEGORY: TRAINING INFORMATION

Mentoring and Learning Support

Each learner gets a mentor who meets them every two weeks. Mentors help with study plans and hard topics. Extra support sessions are available if you fall behind.